

1. Requirements Gathering

1.1 Introduction

This section presents the requirements gathering process conducted at a selected SME that operates in a multi-service environment offering printing services, food (kotas and drinks), and pool table services. The purpose of this phase was to understand the current operational processes, identify existing challenges, and collect user requirements through observation and informal interviews.

1.2 Requirements Gathering Approaches

The requirements were gathered using observation and informal interviews. Observation was used to analyse how daily activities are performed within the business, while interviews with staff members and customers provided deeper insight into operational challenges and user needs.

1.3 Observations

During observation, we noticed that customer orders in the food section are often written on a piece of paper or remembered mentally during less busy periods. This sometimes leads to confusion and misplaced orders. In the photocopy and printing section, customers submit files through multiple channels such as WhatsApp and USB, which are not centrally tracked. Lastly, there's no formal system for tracking pool table usage, which often lead to conflicts among customers regarding usage.

1.4 Interview findings

1.4.1 Kota and Drinks Business

During the interviews, the staff members indicated that customer orders are recorded on paper during busy periods, while orders during less busy periods are sometimes handled from memory. This approach has resulted in occasional loss or mixing of order slips. It was also noted that stock management is informal, as staff rely on visual inspection of available ingredients rather than structured tracking. Certain

ingredients such as Russians, tend to run out unexpectedly due to inconsistent demand patterns.

1.4.2 Photocopy and Printing Services

In the printing section, customers submit files through platforms such as WhatsApp, Bluetooth, and USB devices. However, there is no formal system to track pending or completed print jobs. The Staff also confirmed that no records are kept for completed transactions, and pricing is calculated manually using a calculator, which may lead to inconsistencies.

1.4.3 Pool Table Services

The pool table service operates on a per-game payment model. However, there is no system to track gameplay duration, and customers are allowed to play until one loses. This has led to frequent conflicts, particularly among inexperienced players. Additionally, income from the pool table is not recorded separately from other services.

1.4.4 General Business Challenges

Staff indicated that preparing food orders is time-consuming and managing multiple services simultaneously is difficult without a structured system. The absence of proper record-keeping also makes it challenging to track daily income and overall business performance.

1.4.5 Summary

The requirements gathering process revealed several key issues, including reliance on manual processes, lack of centralized record-keeping, poor tracking of orders and transactions, and absence of performance monitoring. These challenges contribute to operational inefficiencies, customer dissatisfaction, and limited business growth.

2. Requirements Analysis

2.1 Introduction

This section analyses the information gathered to identify key business problems and define system requirements that can improve operational efficiency and service delivery.

2.2 Analysis of the current system

2.2.1 Food and Drinks Services

The food section relies on manual methods to record customer orders, which increases the likelihood of errors such as lost or incorrect orders. Stock management is also informal, resulting in inconsistent availability of ingredients and unexpected shortages.

2.2.2 Printing and Photocopy Operations

The printing section uses multiple input channels without a central tracking system. This creates inefficiencies as print jobs may be lost or delayed. Additionally, the absence of transaction records and manual pricing methods increases the risk of errors.

2.2.3 Pool Table Operations

The pool table service lacks a structured system to track usage or gameplay duration. This results in disputes among customers and makes it difficult to monitor income generated from this service.

2.3 Problem Identification

The analysis identified several major issues affecting the business. The reliance on manual processes leads to inefficiencies and a high risk of human error. The lack of a centralised system limits the ability to track transactions and monitor operations effectively. Poor financial tracking makes it difficult to determine daily income and profitability, while the absence of structured processes negatively impacts customer experience.

2.4 Problem Statement

The SME currently lacks an integrated digital system to manage its operations, resulting in inefficient processes, poor record management, and

limited control over business activities. These challenges reduce service quality, create operational delays, and prevent effective monitoring of performance.

2.5 SYSTEM REQUIREMENTS

2.5.1 Functional Requirements

The system should be capable of capturing and managing food orders to reduce errors and improve efficiency. It should also track printing jobs from submission to completion to ensure accountability. All transactions across services must be recorded to support accurate financial tracking. The system should further manage inventory levels and monitor pool table usage to reduce conflicts.

2.5.2 Non-Functional Requirements

The system should be user-friendly to ensure that staff members can easily operate it without the need for extensive training. It must be reliable to ensure smooth operation without frequent failures. The system should also be efficient in processing transactions quickly to reduce waiting times for customers. Security is important to ensure that business data is protected from unauthorized access. Lastly, the system should be scalable to allow for future improvements and additional features as the business grows.

2.6 Conclusion of the Analysis

The analysis highlights that manual processes and lack of integration negatively impact efficiency, accuracy, and customer satisfaction. A structured digital system is required to improve operations and support better decision-making.